# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years of experience building **C++/Python tooling**, **algorithmic automation**, and reliable systems through graph-based design, simulation, testing, debugging, performance optimization, and cross-functional delivery.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*

- Tripled throughput across 1K+ AI jobs by restructuring **concurrent backend orchestration** with explicit job states, failure isolation, recoverable processing, and observable workflows under production load reliably daily.
- Reduced dashboard latency from 1.2s to 350ms by optimizing **indexed SQL queries** and multi-tenant data models supporting 500+ users with normalized schemas, row-level isolation, and efficient access patterns.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Processed 1K+ files per batch using **Python multithreading**, validating PDF, DOCX, and ZIP inputs while isolating failures and preserving repeatable workflows during concurrent model execution at scale.
- Maintained 90%+ grading consistency across 10K+ submissions through **structured evaluation workflows**, model switching, fallback paths, persistent results, and instructor review controls supporting reproducible verification reliably at scale.

**USC AD Labs** **January 2025 – May 2025**
*ML Research Assistant* *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building an **algorithmic Python pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Raised closed-loop stability 15% through **quantitative verification** of HD maps, ego paths, nearby agents, attention, and distance-aware interactions across diverse autonomous-driving simulations and edge cases repeatedly and reliably.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* *India*

- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports and cross-platform validation across Linux, Windows, and macOS with repeatable nightly workflows at scale.
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while troubleshooting integrations across Windows, Linux, and macOS production environments at scale.

**Pandit Deendayal Energy University** **January 2023 – May 2024**
*Research Fellowship* *India*

- Reduced centralized-control and quantum-risk exposure 25% in benchmark simulations by designing **consensus algorithms** and evaluating five lattice-based cryptographic primitives across distributed protocol scenarios for peer-reviewed research publication.
- Published peer-reviewed **systems research** and presented at IISc Bengaluru, documenting architecture, simulation methodology, threat models, experimental trade-offs, and reproducible technical findings for distributed consensus design and evaluation.

## Skills

**Languages:** C++, Python, Java, Bash, Shell, SQL, JavaScript
**Algorithms & Systems:** Data Structures, Algorithms, Object-Oriented Design, Linux, Concurrency, Multithreading, Process Scheduling, Distributed Systems, System Design
**Testing & Quality:** Unit Testing, Integration Testing, Automated Testing, PyTest, Mockito, Simulation, Debugging, Data Validation, Error Handling
**Tooling & Delivery:** Git, GitHub, GitHub Actions, Jenkins, CI/CD, Docker, Kubernetes, AWS, Azure, GCP
**Data & Engineering:** AWS S3, AWS Glue, ETL Pipelines, Performance Optimization, Technical Documentation

## Projects

**RTL Generation and Connectivity Toolchain - Design Concept**

- Designed a **C++ RTL-generation toolchain** concept parsing Verilog into an abstract syntax tree, resolving connectivity, inserting clock/reset logic, applying deterministic transformations, emitting synthesizable RTL, and running lint checks.

**Clock and Reset Verification Flow - Design Concept**

- Designed a **clock/reset verification flow** concept using Python and Make scripts, graph-based domain analysis, reset-deassertion checks, configurable rule engines, structural reports, regression tests, and synthesis-ready deliverables reliably.

**CPU Scheduling Simulator** | Live

- Implemented **C++ scheduling algorithms** for FCFS, SJF, SRT, Round Robin, priority, and MLQ, with edge-case handling, process-state visualization, and dynamic turnaround and waiting-time calculations across simulations reliably.